Supplementary Derivation of the Affine Delivery-Time Coefficients

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1 Purpose and notation

This note supplies the algebra omitted from the letter for space. In particular, it derives the link-specific coefficients $A^{(l)}$ and $B^{(l)}$ in

$$\overline{T}^{(l,M)} = \sigma\left(A^{(l)} + B^{(l)} M_A^{(l,M)}\right), \tag{1}$$

and then shows how this affine representation yields the continuous airtime-aligned allocation used by the implementation.

The successful-transmission and collision holding times are measured in slots. For device type $g \in \{M, S\}$,

$$\tau_T^{(l,g)} = M_A^{(l,g)} \frac{L_P + L_{MH} + 80}{R^{(l)}\sigma} + T_{OH}^{(l)}, \tag{2}$$

$$\tau_F^{(l)} = \frac{T_{PH,C} + L_R/R_B^{(l)} + \mathrm{DIFS}}{\sigma}. \tag{3}$$

Consequently, the MLD successful-transmission time can be written as

$$\tau_T^{(l,M)} = B^{(l)} M_A^{(l,M)} + T_{OH}^{(l)}, \qquad B^{(l)} = \frac{L_P + L_{MH} + 80}{R^{(l)}\sigma}. \tag{4}$$

The mean successful A-MPDU delivery time is related to the successful A-MPDU rate by

$$\frac{\overline{T}^{(l,M)}}{\sigma} = \frac{1}{\lambda_{\text{out}}^{(l,M)}}. \tag{5}$$

2 Contention-free link

When $n^{(l,S)} = 0$, every MLD attempt succeeds. Including the mean backoff and the attempt slot, the successful A-MPDU rate is

$$\lambda_{\text{out}}^{(l,M)} = \frac{1}{(W^{(l,M)} + 1)/2 + \tau_T^{(l,M)}}. \tag{6}$$

Substituting (4) into (6) and using (5) gives

$$\frac{\overline{T}^{(l,M)}}{\sigma} = \frac{W^{(l,M)} + 1}{2} + T_{OH}^{(l)} + B^{(l)} M_A^{(l,M)}. \tag{7}$$

Thus, for a contention-free link,

$$A^{(l)} = \frac{W^{(l,M)} + 1}{2} + T_{OH}^{(l)}. \tag{8}$$

3 Link coexisting with SLDs

3.1 Eliminating the MLD attempt probability

Fix a link l with $N = n^{(l,S)} > 0$, and abbreviate

$$p_M = p_A^{(l,M)}, \qquad p_S = p_A^{(l,S)}, \qquad q_g = \frac{\lambda_{\text{out}}^{(l,g)}}{\alpha^{(l)} p_A^{(l,g)}}.$$

Here q_g is the steady-state attempt-probability term for a type-g device. The coupled success-probability equations can be written as

$$p_S = (1 - q_M)(1 - q_S)^{N-1}, \tag{9}$$

$$p_M = (1 - q_S)^N. \tag{10}$$

Equation (10) implies

$$1 - q_S = p_M^{1/N}. \tag{11}$$

Using (11) in (9) yields

$$1 - q_M = \frac{p_S}{p_M^{(N-1)/N}} = \frac{p_S p_M^{1/N}}{p_M}, \tag{12}$$

$$q_M = \frac{p_M - p_S p_M^{1/N}}{p_M}. \tag{13}$$

Since $\lambda_{\text{out}}^{(l,M)} = \alpha^{(l)} p_M q_M$, (13) gives the useful identity

$$\lambda_{\text{out}}^{(l,M)} = \alpha^{(l)} \Delta^{(l)}, \qquad \Delta^{(l)} = p_M - p_S p_M^{1/N}. \tag{14}$$

At the nonzero operating point used in the letter, $\Delta^{(l)} > 0$.

3.2 Separating the MLD aggregation-dependent term

For $N > 0$, the reciprocal idle probability in the state-based model is

$$\begin{aligned} \frac{1}{\alpha^{(l)}} &= 1 + \tau_F^{(l)} + \left(\tau_T^{(l,M)} - \tau_F^{(l)}\right) p_M \\ &\quad + N\left(\tau_T^{(l,S)} - \tau_F^{(l)}\right) p_S \left(1 - p_M^{1/N}\right) - \tau_T^{(l,M)} p_S p_M^{1/N}. \end{aligned} \tag{15}$$

The two terms containing $\tau_T^{(l,M)}$ combine as

$$\tau_T^{(l,M)} p_M - \tau_T^{(l,M)} p_S p_M^{1/N} = \tau_T^{(l,M)} \Delta^{(l)}. \tag{16}$$

The remaining collision terms satisfy

$$\tau_F^{(l)} - \tau_F^{(l)} p_M = \tau_F^{(l)} (1 - p_M).$$

Therefore, (15) becomes

$$\frac{1}{\alpha^{(l)}} = \tau_T^{(l,M)} \Delta^{(l)} + C^{(l)}, \tag{17}$$

$$\begin{aligned} C^{(l)} &= 1 + \tau_F^{(l)} (1 - p_M) \\ &\quad + N\left(\tau_T^{(l,S)} - \tau_F^{(l)}\right) p_S \left(1 - p_M^{1/N}\right). \end{aligned} \tag{18}$$

Combining (5), (14), and (17) gives

$$\frac{\overline{T}^{(l,M)}}{\sigma} = \frac{1}{\alpha^{(l)}\Delta^{(l)}}$$
$$= \tau_T^{(l,M)} + \frac{C^{(l)}}{\Delta^{(l)}}. \tag{19}$$

Finally, substituting (4) into (19) produces (1), where

$$A^{(l)} = \frac{1 + \tau_F^{(l)}(1 - p_M) + N\left(\tau_T^{(l,S)} - \tau_F^{(l)}\right) p_S \left(1 - p_M^{1/N}\right)}{p_M - p_S p_M^{1/N}} + T_{OH}^{(l)}, \tag{20}$$

$$B^{(l)} = \frac{L_P + L_{MH} + 80}{R^{(l)}\sigma}. \tag{21}$$

Restoring $N = n^{(l,S)}$, $p_M = p_A^{(l,M)}$, and $p_S = p_A^{(l,S)}$ in (20) gives exactly the coefficient reported in the letter.

The probabilities p_M and p_S are the steady-state solution of the coupled backoff equations. Once that operating point is obtained for a link, all quantities in $A^{(l)}$ are independent of the MLD aggregation variable $M_A^{(l,M)}$. The coefficient $B^{(l)} > 0$ is the incremental airtime, in slots, of one additional MLD MPDU.

4 Continuous airtime-aligned allocation

For a fixed continuous active-link set $\mathcal{I} \subseteq \{1, \ldots, L\}$, equal delivery times make the variance objective zero whenever the corresponding allocations are positive. Let

$$t = \frac{\overline{T}^*}{\sigma}.$$

Using (1), airtime alignment requires

$$A^{(l)} + B^{(l)}\widetilde{M}_A^{(l,M)} = t, \qquad l \in \mathcal{I}. \tag{22}$$

Hence,

$$\widetilde{M}_A^{(l,M)} = \frac{t - A^{(l)}}{B^{(l)}}. \tag{23}$$

Substitution into $\sum_{l\in\mathcal{I}} \widetilde{M}_A^{(l,M)} = W_{BA}$ gives

$$t = \frac{W_{BA} + \sum_{l\in\mathcal{I}} A^{(l)}/B^{(l)}}{\sum_{l\in\mathcal{I}} 1/B^{(l)}}. \tag{24}$$

Equations (23)–(24) are the closed-form continuous solution on $\mathcal{I}$. Starting with all links, any link for which $\widetilde{M}_A^{(l,M)} \leq 0$ is removed from the continuous active set, and the two equations are recomputed on the reduced set. This process terminates after at most L active-set passes because each nonterminal pass removes at least one link.

5 BAW-feasible integer construction

Let $\mathcal{I}$ be the stabilized continuous active set. First set

$$M_A^{(l,M),*} = \begin{cases} \left\lfloor \widetilde{M}_A^{(l,M)} \right\rfloor, & l \in \mathcal{I}, \\ 0, & l \notin \mathcal{I}. \end{cases}$$

The residual number of MPDUs is

$$r = W_{BA} - \sum_{l=1}^{L} M_A^{(l,M),*} \tag{25}$$

$$= \sum_{l \in \mathcal{I}} \left(\widetilde{M}_A^{(l,M)} - \left\lfloor \widetilde{M}_A^{(l,M)} \right\rfloor \right). \tag{26}$$

Because W_{BA} and all floored allocations are integers, r is an integer. Each fractional part lies in $[0, 1)$, so

$$0 \leq r < |\mathcal{I}|.$$

Assigning one additional MPDU to each of the r links with the largest fractional parts therefore constructs a nonnegative integer allocation satisfying

$$\sum_{l=1}^{L} M_A^{(l,M),*} = W_{BA}.$$

A link removed by the continuous active-set procedure receives zero. A link retained in the continuous active set can also receive zero after integer rounding; such a link is inactive in the implemented allocation.

6 Correspondence with the reference implementation

The derivation is implemented directly in `solve.py`. The function `calc_AB` evaluates (8), (20), and (21). The function `solve_closed_form` evaluates (23)–(24), performs the active-set updates, and applies the largest-remainder construction. The public source is available at `https://github.com/chenmengmeng2025/mlo-baw`.